Chapter 02: Literature Survey

Dealing with waste effectively turns into a major worldwide problem these days. Urban areas keep expanding, and industries grow even faster. Old-fashioned ways of sorting waste depend too much on people doing it by hand. Those methods take a lot of effort. They end up being slow and full of mistakes pretty much. Artificial intelligence moves ahead quickly these days. Computer vision tech does the same. Automated systems for classifying waste start to look like a solid fix for all this. In this chapter, we go over what researchers have already studied. We check out open-source efforts too. Plus, we point out the spots that still need work in AI for handling waste.

2.1 Overview Of Relevant Literature

Table 1: research papers

Sno.	Title	Authors	Year	Description
1.	Artificial intelligence for waste management in smart cities: a review	Bingbing Fang, Jiacheng Yu, Zhonghao Chen, Ahmed I. Osman, Mohamed Farghali, Ikko Ihara, Essam H. Hamza, David W. Rooney, Pow-Seng Yap	2023	Reviews AI tools for smart waste management, showing efficiency gains but limited by high cost and lack of uniform data.
2.	Smart waste management: A paradigm shift enabled by artificial intelligence	David B. Olawade, Oluwaseun Fapohunda, Ojima Z. Wada, Sunday O. Usman, Abimbola O. Ige, Olawale Ajisafe, Bankole I. Oladapo	2024	Explores AI techniques for collection, sorting, and recycling; notes benefits with IoT but challenges in data quality and cost.
3.	Smart Waste Segregation and Monitoring System using IoT	V. Sowndharya, P. Savitha, S. Hebziba Jeba Rani	2019	Proposes an IoT-based system for waste segregation and monitoring, limited by sensor accuracy and scalability.

4.	Smart waste management and classification system using advanced IoT and AI technologies	A. Alourani, M. U. Ashraf, M. Aloraini	2025	This setup recommends the use of smart bins that are connected through IoT. They rely on CNN models such as VGG-19, which hits up to 97.7 percent accuracy. The main point here is to handle waste classification on its own.
5.	Smart Waste Segregation System: Proximity-Triggered Waste Classification Using Convolutional Neural Networks	K. Bhowmick, C. Bhadane, P. Shah, H. Patel, C. Lakhani, and R. Pendam	2025	Uses CNN with proximity sensors for automated waste segregation, improving accuracy and reducing manual effort. Limited by dataset size, deployment issues, and sensor dependence.
6.	Smart Waste Segregation System Using Image Processing	Dr. Jyotsna S. Gawai, Khushal R. Bhavsar, Sanchit Shahare	2025	Uses ESP32-CAM and deep learning (MobileNetV2 best at 94.2%) for 5-category waste segregation. Issues include misclassification of organic/glass and ~6% mechanical errors.
7.	IoT-based Intelligent Waste Management System	Ahmed et al.	2023	IoT sensor with LoRa and GSM make collection routes better and helps in cutting costs by monitoring bin fill levels and are focused only on detection and collection, with challenges in increasing scalability and using energy in a good manner
8.	Environmental impacts in e-waste management using deep learning	S. Ramesh et al.	2025	Uses EfficientNet, MobileNet, and Sequential NN for e-waste classification (~98% accuracy). Limited by dataset diversity and lack of industrial-scale testing.

9.	Investigation of a deep learning-based waste recovery framework for sustainability and a clean environment using IoT	A. Khan et al.	2025	Proposes IoT + deep learning (WMR-DL) framework improving waste recovery by ~30%. Needs large-scale municipal validation and energy optimization.
10.	Enhancing waste sorting and recycling efficiency: robust deep learning-based approach for classification and detection	Y. Li et al.	2024/25	Introduces robust CNN models for urban waste sorting with higher accuracy than standard CNNs. Still faces dataset imbalance and high computational cost.
11.	Turning trash into treasure: Exploring the potential of AI in municipal waste management	A. El Jaouhari, A. Samadhiya, A. Kumar, E. Mulat-weldemeskel, R. Kumar	2025	SLR on AI for municipal waste; improves forecasting, sorting, and recycling but limited by data and ethical issues.
12.	A Systematic Review of AI-Based Techniques for Automated Waste Classification	F. Fotovvatikhah, I. Ahmedy, R. Md Noor, M. U. Munir	2025	Reviews ML/DL waste classification; CNN/hybrids >95% accuracy; issues with datasets and standardization.
13.	AI-Driven Innovations in Waste Management: Catalyzing the Circular Economy	Ahmed Snoun, Miratul Khusna Mufida, Abdessamad Ait El-Cadi, Thierry Delot	2025	Highlights AI (ML, DL, IoT, robotics) for waste efficiency; benefits high but faces interoperability and ethics concerns.
14.	The AI-Powered Cleanup: A Revolution in Solid Waste Management	Tanuja Tiwari, Pallavi	2025	AI with IoT, robotics, and analytics improves routing, segregation, and recycling; limited by data gaps and scalability.

15.	IoT-Enabled Waste Management in Smart Cities: A Systematic Literature Review	Moulay Lakbir Tahiri Alaoui, Meryam Belhiah, Soumia Ziti	2025	Reviews IoT-enabled smart bins and networks for route optimization and recycling; faces high cost and interoperability issues.
16.	The Role of AI and ML in Waste Management and Sustainability	Neeraj Arya et al.	2023	AI/ML boost sorting, routing, and monitoring, reducing waste and emissions; challenges are cost and data quality.
17.	AI-Powered Smart Waste Management	Ritika Chaudhary et al.	2024	IoT + AI achieved up to 94% accuracy, cost/fuel savings, and overflow reduction; limited by high cost and data reliance.
18.	Smart Waste Sorting System Using AI	Janani T, Jenifer Barbara J, Swetha K, Shanmugapriya L R, Mrs C Janani	2025	CNN + IoT system reached 95% accuracy with real-time bin monitoring; limited to two categories and lacks robotic sorting.
19.	The Use of Recent Smart Waste Management Systems with AI and Computer Vision	Lakshmi M.	2025	Review of AI + CV for waste management showing efficiency gains in cities; challenges include high costs, generalizability, and ethics.
20.	Literature Review of Automated Waste Segregation System using Machine Learning: A Comprehensive Analysis	Myra G. Flores, Jose B. Tan Jr.	2020	Review identifies CNN as most effective for waste segregation; gaps include bulk waste handling, e-waste focus, and cost issues

A look at recent work reveals an obvious, quick shift towards using deep learning in automating how we sort and classify waste. The aim is simple: build systems that speed up operations, increase accuracy, and become more reliable compared to older manual methods.

At the center of this automation are Convolutional Neural Networks. Active in the field of images, the networks lend themselves well because they inherently learn to pull out hierarchical features from raw images alone: from simple edges up to intricate object details. In the context of this work, CNNs are apt at parsing the messy visuals of mixed waste streams. Already, initial research has shown that tailored architectures achieve high accuracy in classifying the basic categories of paper, plastic, metal, and glass.

Yet, there's a big leap from sterile lab tests to deployment in the real world. Inconsistent lighting in real environments, together with busy backdrops, partly occluded objects, and enormous intra-class variation—a plastic bottle may come in many shapes and conditions—severely tests how robust these models are.

In recent times, most deployment efforts towards this problem have been highly reliant on transfer learning. This minimizes the necessity of large volumes of specialized data or expensive computation by reusing networks which are pre-trained on large image datasets like ImageNet. Fine-tuning strong, pre-trained models on smaller image sets of waste will easily adapt the learned features to the new domain. The upside of transfer learning is obvious: faster training, less data, and usually much better generalization. We had various pre-trained architectures but ResNet50, MobileNetV2, and InceptionV3 came on top time and again for their solid performance on waste classification tasks. InceptionV3 was the best among the final shortlisted ones as it processes information at a number of scales within a single block, this helps tease apart subtle differences between visually similar items of waste.

2.2 Key Gaps In The Literature

1. Bhowmick et al. (2018): Limited to 5 waste categories, misclassification between organic & glass, ~6.2% mechanical failure.

2. Patil et al. (2019): They only focused on metallic waste and the dataset which they used was very small and was not scalable.

3. Mohan et al. (2019): Segregation is limited to dry and wet waste only and it also does not handle mixed or e-wastes and it requires sensor calibration which is quite an hefty task.

4. Joshi & Saxena, 2019 : Works only in controlled environments. Their implementation does not understand light and background change

5. Yadav & Deshpande (2019): it declines the for items not trained on small extent of categories and their need for sensors actually raise privacy related issues.

6. Shinde et al. (2019): Handles only 3 categories, highly dependent upon the calibration of sensors, not suitable for large-scale deployment.

7. Chaudhari et al. (2020)-Accuracy falls to low with unsegregated and mixed waste and they require high computing power.

8. Prasad et al. (2020): Trained on a limited dataset; generalization is not appropriate on unseen waste items, and absence of real-world validation.

9. Khan et al. (2020)-It requires continuous internet and power supply, and is also limited to urban smart-bin setups.

10. Flores & Tan (2020): CNN best for segregation, but lacks cost-effective deployment & e-waste focus.

11. Ali et al. (2021): The system is quite effective in smart cities but has a problem or limitation of cost for infrastructure and integration with policy.

12. Patil & Kulkarni, 2021: This detects plastic well but just ignores biodegradable/metal waste; it is not scalable.

13. Shukla et al. (2021): Works only in lab setups and is not quite good for large-scale variation in waste.

14. Sharma et al. (2021):it has high accuracy but is only limited to dry waste & wet waste it also requires advanced image datasets.

15. Kumar et al. (2022): The performance of this model decreases by a lot on real-world data sets and it was not scalable also the cost of training this model is too high .

16. Verma et al. (2022) – Have focused on the segregation and recycling of plastics. Further, it lacks multi-category segregation and real-time deployment.

17. Lakshmi 2025: Their solution has high infrastructure costs and has some ethical issues also this models lack generalizability across cities.

18. Janani et al. (2025): CNN and IoT system was implemented however this model of their is only limited to 2 categories and provides no robotic sorting.

19. Flores & Tan (2020) [Review]: Identifies CNN as effective, but highlights gaps in bulk waste, e-waste, and cost barriers.

20. Update 2025 Additional Review: AI and CV improves efficiency by a margin but yet it needs further cost reduction because it is too costly it also needs scalable deployment and stronger privacy policies for safety of the user.